

A nearby norming point gives an almost-norming evaluation

MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby

theorem

Two triangle inequalities transfer absolute norm attainment to a unit vector.

Statement

Let $n \in \mathbb{N}$ and use reference coordinates $V = \mathbb{R}^n$ with the sup norm. Let $p : V \rightarrow \mathbb{R}$ be a continuous seminorm with specified constants $a_p, b_p > 0$ such that

$$a_p \|x\|_\infty \leq p(x) \leq b_p \|x\|_\infty \quad (x \in V).$$

The lower comparison makes p a norm. Let $r : V \rightarrow \mathbb{R}$ be continuous and real linear, with $|r(y)| \leq p(y)$ for every $y \in V$ and let $x, z \in V$, $\delta \in \mathbb{R}$ satisfy

$$p(x) = 1, \quad p(x - z) \leq \delta, \quad |r(z)| = p(z).$$

Then $1 - 2\delta \leq |r(x)|$.

Assumptions

The four hypotheses are contraction $|r(y)| \leq p(y)$ for all y , the unit condition at x , the error bound at $x - z$, and absolute norm attainment at z . No independent $\delta \geq 0$ or $\delta > 0$ hypothesis is imposed; nonnegativity follows from $0 \leq p(x - z) \leq \delta$.

Conclusion

The same functional r gives $|r(x)| \geq 1 - 2\delta$. It norms z exactly and evaluates the nearby unit vector x approximately; these points are not interchanged.

Proof route

Compare the norm at the nearby point with the unit norm, then bound the change in the same linear functional.

▼ Proof steps

1. The seminorm triangle inequality gives $1 = p(x) \leq p(x - z) + p(z)$, hence $1 - \delta \leq p(z)$. By linearity and contraction,

$$p(z) = |r(z)| = |r(x) - r(x - z)| \leq |r(x)| + |r(x - z)| \leq |r(x)| + \delta.$$

Combining the two displayed bounds yields $1 - 2\delta \leq |r(x)|$. This also covers $\delta = 0$ and estimates whose lower bound is negative.

The contraction is evaluated at the difference of the two displayed points; the estimate is an elementary seminorm calculation.

Main citations

- Exact declaration and proof — `MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby`

Lean source signature (exact)

```
theorem abs_eval_lower_of_norms_nearby {n : ℕ} (M : NormModel n)
  {r : Coord n →L[ℝ] ℝ} (hr : IsDualContraction M r)
  {x z : Coord n} (hx : M.p x = 1)
  {δ : ℝ} (hδ : M.p (x - z) ≤ δ)
  (hnorm : |r z| = M.p z) :
  1 - 2 * δ ≤ |r x|
```

The four named proof arguments are precisely the four mathematical hypotheses. The final inequality is the conclusion.

In the source	Mathematical meaning
<code>Coord n</code>	The reference space $V = \mathbb{R}^n$ with its sup norm; <code>Coord n</code> abbreviates $\text{Fin } n \rightarrow \mathbb{R}$. Lean uses indices $0, \dots, n-1$ and the formulas use $1, \dots, n$.
<code>M : NormModel n; M.p</code>	The input <code>M</code> contains the continuous seminorm p and constants $a_p, b_p > 0$ with $a_p \ x\ _\infty \leq p(x) \leq b_p \ x\ _\infty$ for every x . <code>M.p x</code> is the scalar $p(x)$.
<code>r : Coord n →L[ℝ] ℝ; hr : IsDualContraction M r</code>	The given $r : V \rightarrow \mathbb{R}$ is continuous and real linear, with $ r(y) \leq p(y)$ for every $y \in V$.
<code>hx : M.p x = 1</code>	The point x at which we want an approximate evaluation satisfies $p(x) = 1$.
<code>hδ : M.p (x - z) ≤ δ</code>	The nearby point z satisfies $p(x - z) \leq \delta$, with $\delta \in \mathbb{R}$.
In the source	Mathematical meaning
<code>hnorm : r z = M.p z</code>	The same r attains the norm at z in absolute value: $ r(z) = p(z)$.
<code>1 - 2 * δ ≤ r x </code>	The conclusion $1 - 2\delta \leq r(x) $ is an evaluation at x , not at z .

$M.p \ (x - z)$ evaluates the norm on a difference; $r \ x$ evaluates the given functional. Braces permit inference of r, x, z, δ , without changing their mathematical roles.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 13–14:

```
private abbrev Coord (n : ℕ) := Fin n → ℝ
private abbrev NormModel (n : ℕ) := EquivalentSeminorm (Coord n)
```

Exact source lines 21–21:

```
private abbrev IsDualContraction {n : ℕ} (M : NormModel n) (r : Coord n → L[ℝ] ℝ) := ∀ x, |r x| ≤ M.p x
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 7–55. The full original context is retained with the source evidence.

▼ Exact content identity

Declaration: MathlibAnnex.Satellite.abs_eval_lower_of_norms_nearby

Accepted content SHA-256: 76a978dcc00e16703cf5650212cdf371c6a0d7983de05960d47b27b416cfd983

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Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73